

YINGJIN LI

M.Phil. Student, The Hong Kong University of Science and Technology (Guangzhou)

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Research Interests: Visual Anomaly Detection · Open-World Visual Understanding
Multimodal Learning · Vision-Language Models

Education

The Hong Kong University of Science and Technology (Guangzhou)

M.Phil., Data Science and Analytics

Sep 2025 - Jun 2027 (expected)

Thesis Supervisor: Prof. Lei Zhu

Recipient of the **Red Bird Scholarship (RMB 240,000)**.

Wuhan Textile University

B.Eng., Data Science and Big Data Technology

Sep 2021 - Jun 2025

Average score: 86.7/100 · Academic ranking: **Top 5%** · Overall ranking: **1st in major**

Research & Publications

ROAD: Rule-Grounded Context-Aware Open-World Driver Anomaly Detection

First Author · *Advances in Neural Information Processing Systems (NeurIPS)*, 2026 · **Accepted**

Apr 2026 - Present

- Designed and implemented a **multimodal anomaly detection framework** that integrates driver video, driving context, and explicit behavioral rules to identify previously unseen driver anomalies.
- Developed **rule-grounded reasoning** that translates domain knowledge into structured guidance for multimodal large language models, supporting explanations of why a behavior is abnormal beyond predefined action categories.
- Integrated **context-aware visual reasoning** to distinguish visually similar behaviors across driving situations and account for context-dependent safety rules in open-world anomaly understanding.
- Built an evaluation pipeline across driver-monitoring datasets with free-form anomaly descriptions, human assessments, and MLLM baselines to assess open-world generalization and model-human agreement.

PriSM-AD: Learning Common Normal Priors and Anomaly Semantics for Diffusion-Based Anomaly Detection

First Author · *IEEE Transactions on Multimedia (TMM)* · **Under Review**

- Designed a **two-stage diffusion framework** that separates learning common normal visual priors from anomaly localization, combining normality modeling with semantic cues for industrial defect detection.
- Developed **prior-guided reconstruction** to recover normal appearance while suppressing anomalous structures, providing a normal reference for identifying subtle defects.
- Integrated **anomaly semantics and reconstruction cues** to support both image-level anomaly detection and pixel-level localization across industrial categories.
- Analyzed reconstruction quality, cross-category generalization, anomaly localization, and inference efficiency.

Fashion Image Retrieval Based on Multimodal Features Enhancement and Fusion

Yingjin Li, Shufan He, Zhaojing Wang, Jin Huang, Xinrong Hu, and Li Li.

First Author · *Computer Animation and Social Agents (CASA 2024)*.

Springer, CCIS 2374, pp. 118-131, 2025. doi.org/10.1007/978-981-96-2681-6_9

Honors & Awards

- National Scholarship**, Ministry of Education of China, 2024
- First Prize, Hubei Province**, China Mathematical Contest in Modeling
- National First Prize**, Teddy Cup Data Mining Challenge
- National First Prize**, Huazhong Cup Mathematical Modeling Challenge